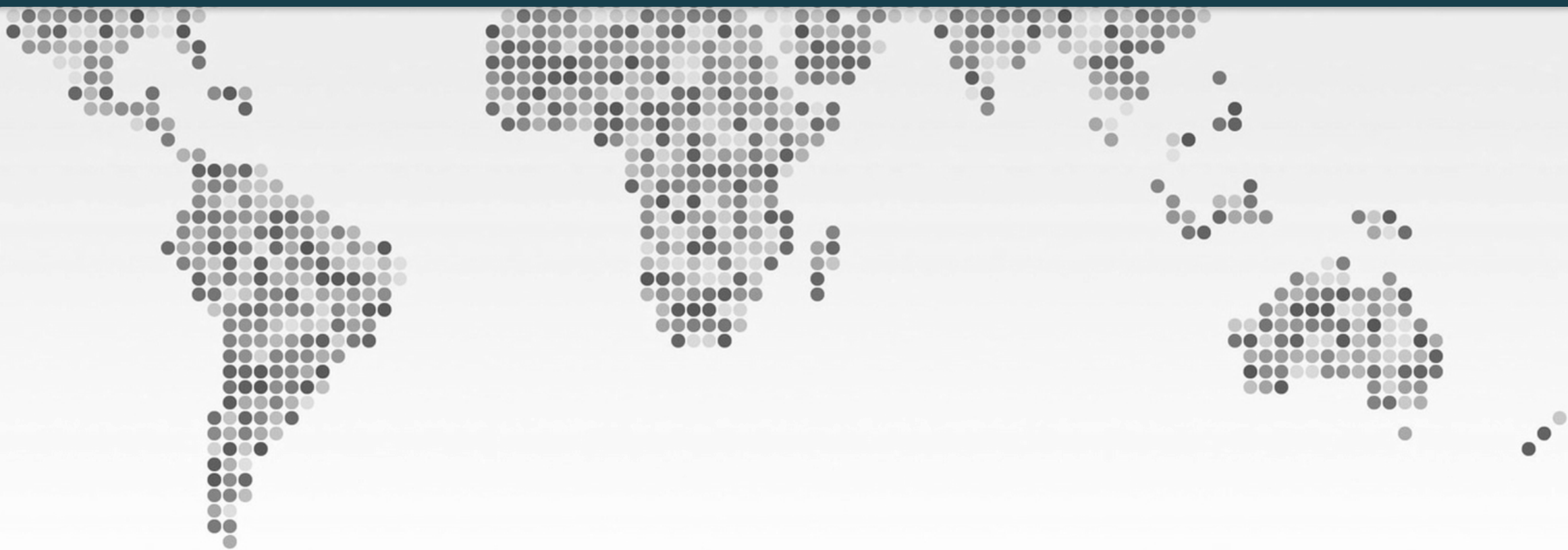


Software Engineering for Artificial Intelligence

Project Idea Discussion



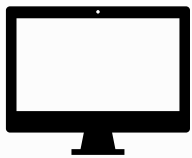
Administrative Information - Exams and Grades

The final grade will be a weighted sum of the grades received to three parts:



The oral exam will consist of 3/4 questions will be asked, for a duration up to one hour (this is not often the case, but it happened in the past).

40%



The project will make you able to engineer an AI-enabled system, experimenting monitoring tools and implementing responsible AI approaches.

60%

$$\rightarrow G = 0.4G_{\text{oral}} + 0.6G_{\text{project}}$$

The project development is NOT mandatory: In this case, the grade will be the combination of the first two grades $\rightarrow$ the final evaluation will be 24, at most.

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- Clearly, a public **repository** containing all the code and raw results of the project must be submitted. Mindfully using **GitHub** is necessary to demonstrate each specific contribution to the project.
- At the end of the project. You will be asked to deliver a **presentation** of the work you have done. More details on this later on!

List of topics

Social aspects of SE for AI development communities

- Analyze open source communities to understand how SE and AI practitioners interact and communicate
- Design and evaluate AI community-specific socio-technical metrics

List of topics

Social aspects of SE for AI development communities

SE for LLMs- and Prompt-Engineering

- Study the evolution of prompts in Agent based repositories to understand what changes and why
- Evaluate LLMs consistency over multiple runs for SE tasks
- Understand how prompt quality evolves over multiple interactions between humans and LLMs
- Governance of AI agents: regulations and AI Act
- Engineering coding agents through Context Files

List of topics

Social aspects of SE for AI development communities

SE for LLMs- and Prompt-Engineering

SE practices in AI- and Agent-enabled systems

- Evaluate how AI and Agents are tested (or test) in the wild
- Understand specific MLOps practices for Agent-enabled systems

List of topics

Social aspects of SE for AI development communities

SE for LLMs- and Prompt-Engineering

SE practices in AI- and Agent-enabled systems

Ethics and Sustainability of AI- and Agent-enabled systems

- Empirically evaluate fairness metrics for natural language and image generation models
- Evaluate ethical practices of using and engineering agents in SE
- Evaluate environmental sustainability aspects of using agents for SE tasks (and the trade-offs with other non functional requirements)

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Ethics and Sustainability of AI- and Agent-enabled systems

Technical debt in AI- and Agent-enabled systems

- Code smells and other quality aspects of AI generated code
- Self-admitted and unadmitted technical debt in AI generated code

Project Idea Discussion

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Technical debt in AI- and Agent-enabled systems

**Anything close or related to these topics
is fine, as long as you discuss it with us!**

MANDATORY

Register your group,
indicating the project idea,
by **March 22, 2026**.



<https://forms.gle/RVTKCyi7JmHRzXkU8>